

Solutions: Additional Exercises 1

Positive Series

Series and Integral Transforms

Exercise 1 — Finding Exact Sums

(a) ★ $\sum_{n=1}^{\infty} \left(\frac{3}{4}\right)^n$

This is a **geometric series** with first term $a = \frac{3}{4}$ and ratio $r = \frac{3}{4}$. Since $|r| < 1$, the series converges and

$$\sum_{n=1}^{\infty} \left(\frac{3}{4}\right)^n = \frac{a}{1-r} = \frac{\frac{3}{4}}{1-\frac{3}{4}} = \frac{\frac{3}{4}}{\frac{1}{4}} = \boxed{3}.$$

(b) ★ $\sum_{n=0}^{\infty} \frac{2^n + 3^n}{5^n}$

Split the series using linearity:

$$\sum_{n=0}^{\infty} \frac{2^n + 3^n}{5^n} = \sum_{n=0}^{\infty} \left(\frac{2}{5}\right)^n + \sum_{n=0}^{\infty} \left(\frac{3}{5}\right)^n.$$

Both are geometric series (starting at $n = 0$) with $|r| < 1$:

$$= \frac{1}{1-\frac{2}{5}} + \frac{1}{1-\frac{3}{5}} = \frac{5}{3} + \frac{5}{2} = \frac{10}{6} + \frac{15}{6} = \boxed{\frac{25}{6}}.$$

(c) ★ $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$

Use **partial fractions**: $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$.

The partial sum **telescopes**:

$$S_N = \sum_{n=1}^N \left(\frac{1}{n} - \frac{1}{n+1}\right) = 1 - \frac{1}{N+1} \xrightarrow{N \rightarrow \infty} \boxed{1}.$$

(d) ★ $\sum_{n=1}^{\infty} \frac{1}{n(n+2)}$

Partial fractions: $\frac{1}{n(n+2)} = \frac{1}{2} \left(\frac{1}{n} - \frac{1}{n+2} \right)$.

Telescoping with step 2:

$$S_N = \frac{1}{2} \left(1 + \frac{1}{2} - \frac{1}{N+1} - \frac{1}{N+2} \right) \xrightarrow{N \rightarrow \infty} \frac{1}{2} \cdot \frac{3}{2} = \boxed{\frac{3}{4}}.$$

(e) ★★ $\sum_{n=1}^{\infty} \left[\sin\left(\frac{1}{n}\right) - \sin\left(\frac{1}{n+1}\right) \right]$

Define $b_n = \sin(1/n)$. The general term is $b_n - b_{n+1}$, which **telescopes**:

$$S_N = b_1 - b_{N+1} = \sin(1) - \sin\left(\frac{1}{N+1}\right) \xrightarrow{N \rightarrow \infty} \sin(1) - 0 = \boxed{\sin 1}.$$

(f) ★★ $\sum_{n=2}^{\infty} \ln\left(1 - \frac{1}{n^2}\right)$

Factor the argument: $1 - \frac{1}{n^2} = \frac{(n-1)(n+1)}{n^2}$. Using log rules:

$$\ln\left(\frac{(n-1)(n+1)}{n^2}\right) = \ln(n-1) - \ln n + \ln(n+1) - \ln n.$$

Write $a_n = \ln(n-1) - \ln n$ and $c_n = \ln(n+1) - \ln n$. Then:

$$S_N = \sum_{n=2}^N a_n + \sum_{n=2}^N c_n.$$

The first sum telescopes to $\ln 1 - \ln N = -\ln N$. The second telescopes to $\ln(N+1) - \ln 2$. Hence

$$S_N = -\ln N + \ln(N+1) - \ln 2 = \ln\left(\frac{N+1}{N}\right) - \ln 2 \xrightarrow{N \rightarrow \infty} 0 - \ln 2 = \boxed{-\ln 2}.$$

(g) ★ $\sum_{n=1}^{\infty} \frac{4^{n+1}}{5^n}$

Write $\frac{4^{n+1}}{5^n} = 4 \cdot \left(\frac{4}{5}\right)^n$. Geometric series:

$$4 \sum_{n=1}^{\infty} \left(\frac{4}{5}\right)^n = 4 \cdot \frac{\frac{4}{5}}{1 - \frac{4}{5}} = 4 \cdot \frac{\frac{4}{5}}{\frac{1}{5}} = 4 \cdot 4 = \boxed{16}.$$

(h) ★★ $\sum_{n=0}^{\infty} \frac{1}{(3n+1)(3n+4)}$

Partial fractions: $\frac{1}{(3n+1)(3n+4)} = \frac{1}{3} \left(\frac{1}{3n+1} - \frac{1}{3n+4} \right)$.

Telescoping (the terms pair up with step 3):

$$S_N = \frac{1}{3} \left(\frac{1}{1} - \frac{1}{3N+4} \right) \xrightarrow{N \rightarrow \infty} \frac{1}{3} \cdot 1 = \boxed{\frac{1}{3} \cdot \frac{1}{1}}.$$

Wait — at $n = 0$ the first term is $\frac{1}{3}(\frac{1}{1} - \frac{1}{4})$, at $n = 1$ it is $\frac{1}{3}(\frac{1}{4} - \frac{1}{7})$, etc. The telescope collapses to

$$S_N = \frac{1}{3} \left(1 - \frac{1}{3N+4} \right) \rightarrow \frac{1}{3}.$$

But note the first factor at $n = 0$ is $3(0) + 1 = 1$ and $3(0) + 4 = 4$, so indeed $S = \boxed{\frac{1}{3}}$.

Correction check: At $n = 0$: $1/(1 \cdot 4) = 1/4$. At $n = 1$: $1/(4 \cdot 7)$. The telescope gives $S = \frac{1}{3}(1/1) = 1/3$. ✓

Exercise 2 — Convergence Tests

(a) ★ $\sum \frac{1}{n^2 + 1}$

Since $\frac{1}{n^2 + 1} \leq \frac{1}{n^2}$ and $\sum \frac{1}{n^2}$ converges (p -series, $p = 2 > 1$), by the **Comparison Test** the series **converges**.

(b) ★ $\sum \frac{n}{2^n}$

Ratio Test: $\frac{a_{n+1}}{a_n} = \frac{n+1}{2^{n+1}} \cdot \frac{2^n}{n} = \frac{n+1}{2n} \rightarrow \frac{1}{2} < 1$. **Converges.**

(c) ★ $\sum \frac{n^2}{n!}$

Ratio Test: $\frac{(n+1)^2}{(n+1)!} \cdot \frac{n!}{n^2} = \frac{(n+1)^2}{(n+1)n^2} = \frac{n+1}{n^2} \rightarrow 0 < 1$. **Converges.**

(d) ★★ $\sum \frac{1}{n(\ln n)^2}, n \geq 2$

Integral Test: $f(x) = \frac{1}{x(\ln x)^2}$ is positive and decreasing.

$$\int_2^{\infty} \frac{dx}{x(\ln x)^2} = \left[-\frac{1}{\ln x} \right]_2^{\infty} = \frac{1}{\ln 2} < \infty.$$

Converges.

(e) ★ $\sum \frac{2^{1/n}}{3^n}$

Since $2^{1/n} \rightarrow 1$, we have $\frac{2^{1/n}}{3^n} \sim \frac{1}{3^n}$ as $n \rightarrow \infty$. By **Limit Comparison** with $\sum (1/3)^n$ (geometric, converges): $L = \lim \frac{2^{1/n}/3^n}{1/3^n} = 1$. **Converges**.

(f) ★ $\sum \frac{\arctan(1/n)}{n}$

Since $\arctan(1/n) \sim 1/n$ as $n \rightarrow \infty$, the terms behave like $1/n^2$. **Limit Comparison** with $\sum 1/n^2$: $L = \lim n^2 \cdot \arctan(1/n)/n = \lim n \arctan(1/n) = 1$. **Converges**.

(g) ★★★★★ $\sum \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{(2n)!}$

Ratio Test:

$$\frac{a_{n+1}}{a_n} = \frac{1 \cdot 3 \cdots (2n-1)(2n+1)}{(2n+2)!} \cdot \frac{(2n)!}{1 \cdot 3 \cdots (2n-1)} = \frac{2n+1}{(2n+2)(2n+1)} = \frac{1}{2n+2} \rightarrow 0 < 1.$$

Converges.

(h) ★★ $\sum \left(\frac{n}{3n-1} \right)^{2n-1}$

Root Test: $a_n^{1/n} = \left(\frac{n}{3n-1} \right)^{(2n-1)/n} \rightarrow \left(\frac{1}{3} \right)^2 = \frac{1}{9} < 1$. **Converges.**

(i) ★ $\sum \left(\frac{n+1}{2n-1} \right)^n$

Root Test: $a_n^{1/n} = \frac{n+1}{2n-1} \rightarrow \frac{1}{2} < 1$. **Converges.**

(j) ★★ $\sum \frac{n!}{1 \cdot 3 \cdot 5 \cdots (2n-1)}$

Ratio Test:

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)!}{1 \cdot 3 \cdots (2n+1)} \cdot \frac{1 \cdot 3 \cdots (2n-1)}{n!} = \frac{n+1}{2n+1} \rightarrow \frac{1}{2} < 1.$$

Converges.

(k) ★★★ $\sum \frac{(2n)!}{4^n (n!)^2}$

Ratio Test: $\frac{a_{n+1}}{a_n} = \frac{(2n+2)!}{4^{n+1}((n+1)!)^2} \cdot \frac{4^n (n!)^2}{(2n)!} = \frac{(2n+2)(2n+1)}{4(n+1)^2} = \frac{2n+1}{2(n+1)} \rightarrow 1.$

Inconclusive.

By **Stirling's approximation**, $(2n)! \sim \sqrt{4\pi n}(2n/e)^{2n}$ and $n! \sim \sqrt{2\pi n}(n/e)^n$, giving:

$$\frac{(2n)!}{4^n (n!)^2} \sim \frac{1}{\sqrt{\pi n}}.$$

Since $\sum 1/\sqrt{\pi n}$ diverges (p -series, $p = 1/2 \leq 1$), the original series **diverges** by Limit Comparison.

Exercise 3 — Theoretical Questions

(a) ★ Necessary condition

The **Necessary Condition for Convergence** states: if $\sum a_n$ converges, then $a_n \rightarrow 0$.

- $\sum \frac{n^2}{n^2+1}$: here $a_n = \frac{n^2}{n^2+1} \rightarrow 1 \neq 0$. **Diverges.**
- $\sum (-1)^n$: here $a_n = (-1)^n$ does not converge to 0. **Diverges.**

(b) ★ Absolute convergence and scalar multiples

Given $\sum a_n$ absolutely convergent and $c > 0$. Then $|ca_n| = c|a_n|$, so $\sum |ca_n| = c \sum |a_n| < \infty$. Hence $\sum ca_n$ is also absolutely convergent.

(c) ★★ Divergence of the sum when one part diverges

Suppose $\sum a_n$ converges and $\sum b_n$ diverges. We claim $\sum (a_n + b_n)$ diverges.

Proof by contradiction: assume $\sum (a_n + b_n)$ converges. Then by linearity of convergent series, $\sum b_n = \sum (a_n + b_n) - \sum a_n$ would also converge — a contradiction. Therefore $\sum (a_n + b_n)$ **diverges**. $\square$

(d) ★★ Bouncing ball

A ball is dropped from height $h = 4$ m, with bounce ratio $r = 2/3$.

Total distance = $h + 2hr + 2hr^2 + \dots = h + \frac{2hr}{1-r}$ (geometric, $|r| < 1$):

$$= 4 + \frac{2 \cdot 4 \cdot \frac{2}{3}}{1 - \frac{2}{3}} = 4 + \frac{\frac{16}{3}}{\frac{1}{3}} = 4 + 16 = \boxed{20 \text{ m}}.$$

(e) ★★ $\sum_{n=0}^{\infty} \tan^n(x)$

This is a geometric series with ratio $r = \tan x$. It converges if and only if $|\tan x| < 1$, i.e., $x \in \left(-\frac{\pi}{4} + k\pi, \frac{\pi}{4} + k\pi\right)$, $k \in \mathbb{Z}$. When it converges, the sum is:

$$S = \frac{1}{1 - \tan x}.$$

Exercise 4 — Integral Test Verification

(a) ★ $\sum_{n=2}^{\infty} \frac{1}{n \ln n}$

$f(x) = \frac{1}{x \ln x}$ is positive and decreasing on $[2, \infty)$.

$$\int_2^{\infty} \frac{dx}{x \ln x} = \left[\ln(\ln x) \right]_2^{\infty} = +\infty.$$

By the **Integral Test**, the series **diverges**.

(b) ★ $\sum_{n=1}^{\infty} \frac{1}{n^2 + 1}$

$f(x) = \frac{1}{x^2 + 1}$ is positive and decreasing.

$$\int_1^{\infty} \frac{dx}{x^2 + 1} = \left[\arctan x \right]_1^{\infty} = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4} < \infty.$$

By the **Integral Test**, the series **converges**.

(c) ★★ $\sum_{n=1}^{\infty} n e^{-n^2}$

$f(x) = x e^{-x^2}$ is positive and decreasing on $[1, \infty)$.

$$\int_1^{\infty} x e^{-x^2} dx = \left[-\frac{1}{2} e^{-x^2} \right]_1^{\infty} = 0 + \frac{1}{2e} = \frac{1}{2e} < \infty.$$

By the **Integral Test**, the series **converges**.

Key takeaways: Geometric series and telescoping are exact-sum workhorses. The p -series criterion ($p > 1$ converges), Ratio/Root tests handle factorial and exponential terms, and the Integral Test bridges series to calculus.